

James Wyckoff

Senior Data Engineer

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Summary

Senior Data Engineer with over **13 years of experience** architecting and building robust data platforms and full-stack solutions across various domains, including **healthcare, fintech, and SaaS**.

Adept at designing, developing, and optimizing **ETL/ELT pipelines**, leveraging **Apache Kafka, Apache Spark, and Databricks** to process and manage large datasets. Highly skilled in cloud technologies such as **Azure, AWS** and **GCP**, and proficient in database management, data modeling, and creating real-time data processing solutions.

Strong track record in improving system performance, reducing latency, and enabling data-driven decision-making. Experienced in mentoring and leading cross-functional teams to deliver scalable, secure, and efficient data architectures. Passionate about continuous learning and driving innovation in data engineering practices.

Skills

- **Languages:** Python (Pandas, NumPy), SQL, Pyspark, Scala, Java, PL/SQL
- **Cloud Platforms:** Azure (Synapse, Data Factory, Databricks, Blob Storage, Kubernetes), AWS (Redshift, Glue, S3, Kinesis)
- **Big Data Tools:** Apache Spark, Apache Kafka, Apache Airflow, Snowflake
- **Databases:** PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Elasticsearch
- **Data Modeling:** Star Schema, Snowflake Schema, Lakehouse Architecture, Data Vault
- **BI & Visualization:** Power BI, Tableau
- **Tools & Practices:** Git, Terraform, Jenkins, Azure DevOps, CI/CD
- **AI/ML:** ML-Ops, Feature Engineering, AI Model Integration
- **DevOps & Automation:** Kubernetes, Azure EKS, Terraform

Experience

Senior Data Engineer

Cencora | Conshohocken, PA

Nov 2021 – Present

- Led the architecture and development of **real-time ETL pipelines** using **Apache Kafka** to integrate and process data from EHR systems, medical devices, and patient portals, ensuring seamless data flow into a **centralized Azure Data Lake** for analytics.
- Architected and implemented a robust **Azure Synapse** data warehouse solution, centralizing clinical, operational, and financial data from disparate sources, providing a unified platform for comprehensive reporting and advanced analytics.
- Utilized **Pandas** and **NumPy** to clean and preprocess large healthcare datasets, such as **EHR data**, **lab results**, and **insurance claims**, ensuring that they were optimized for accurate and timely analysis.
- Developed and deployed **Power BI dashboards** to track critical healthcare KPIs including patient outcomes, resource utilization, and financial performance, helping senior leadership make informed decisions.
- Streamlined backend processes by deploying **microservices** with **Azure EKS** and **Kubernetes**, which improved system scalability, reduced downtime, and ensured high availability for critical healthcare applications.
- Ensured compliance with healthcare regulations by implementing robust **monitoring and logging** using **Azure Monitor** and **CloudTrail**, proactively identifying and resolving any performance issues before they impacted system performance.
- Led the establishment of **CI/CD pipelines** using **Azure DevOps**, automating the entire code deployment process, significantly improving workflow efficiency and reducing deployment errors.
- Successfully reduced patient record retrieval time by **50%** and improved operational efficiency by **30%**, directly enhancing the user experience for both healthcare providers and patients.
- Collaborated with cross-functional teams to design **data governance frameworks**, establishing standards for data security, lineage, and privacy to ensure compliance with **HIPAA** and other relevant regulations.
- Provided mentorship to junior engineers, sharing best practices for **data pipeline optimization**, **cloud infrastructure design**, and **healthcare data integration**, fostering a culture of continuous improvement across the team.
- Drove the adoption of **Agile** and **DevOps** practices, enhancing project delivery timelines and ensuring that all team members were aligned on key goals and deliverables.

Data Engineer

Oracle | Austin, TX

Feb 2017 – Oct 2021

- Spearheaded the design and implementation of **real-time data pipelines** using **Apache Airflow** and **Kafka**, which facilitated the integration and processing of data from multiple internal and external sources, allowing for faster decision-making and data-driven insights across Adobe platforms.

- Optimized large-scale **SQL queries** to reduce query latency and improve system performance, enabling faster data access for analytical applications and reporting tools.
- Developed and managed **CI/CD pipelines** using **Azure DevOps**, streamlining code integration, testing, and deployment processes, which led to a **20% reduction in deployment time** and a significant improvement in code quality.
- Collaborated with **product managers** and **data scientists** to develop scalable solutions that met business needs, integrating complex data sources into a single, unified data platform.
- Led the development of a cloud-based **Azure Databricks** solution, improving the efficiency of data transformation and allowing for faster, more reliable analytics on large datasets using **Apache Spark**.
- Championed the integration of **real-time data processing** and **event-driven architectures** into Adobe's existing data infrastructure, which allowed for more responsive and scalable analytics and reporting.
- Mentored junior engineers, guiding them in the best practices for **data pipeline architecture**, **cloud-based infrastructure**, and **Agile project management**, resulting in a more skilled and productive team.
- Drove performance improvements by implementing **data partitioning**, **caching strategies**, and **query optimization** techniques that led to faster data processing and enhanced analytics performance.
- Played a key role in redesigning Adobe's data architecture, helping to shift the company toward a more **cloud-native** infrastructure, which improved scalability and reduced operational overhead.
- Provided leadership and technical direction on **data architecture** and **cloud strategy**, ensuring Adobe remained at the forefront of data-driven innovation within the creative industry.
- Led the migration of legacy data systems to modern **cloud-based** solutions, ensuring minimal disruption to ongoing operations while improving the long-term sustainability of Adobe's data infrastructure.

Data Engineer/Data Analyst

Nutanix | San Jose, CA

Sep 2014 – Jan 2017

- Built and maintained **real-time data pipelines** using **RabbitMQ** and **Apache Airflow** to facilitate communication between trading engines, risk management systems, and back-office services.
- Developed and optimized **ETL processes** to extract, transform, and load financial data from transactional systems and external market data sources into **Amazon Redshift**, enabling fast and efficient analytics.
- Designed **PostgreSQL schemas** to store transactional and customer data, ensuring efficient data retrieval and seamless integration with external systems.
- Created **real-time dashboards** using **Tableau** to provide key stakeholders with actionable insights into market trends, trade performance, and risk metrics, improving decision-making for trading and risk teams.
- Implemented **data validation** and **monitoring systems** to ensure the integrity and accuracy of financial data, reducing errors and improving data quality.
- Worked with product and business teams to design and implement **custom data solutions**, improving backend services and enhancing the overall functionality of trading and portfolio management systems.

- Played a key role in improving system performance by **optimizing SQL queries** and leveraging **caching strategies**, resulting in a **20% reduction** in page load times and better user experience.
- Assisted in the design and implementation of data models for trading data, portfolio management, and transaction history, improving the scalability and efficiency of the platform.
- Built and maintained automated **data extraction** tools, which significantly improved the data integration process and reduced manual workload.
- Helped reduce the time needed for regulatory reporting by automating the process of **data extraction** and **report generation**, improving the speed and accuracy of compliance efforts.

Junior Data Engineer

Nutanix | San Jose, CA

Sep 2012 – Aug 2014

- Designed and implemented **ETL pipelines** to automate the processing of educational assessment data, ensuring seamless data flow from multiple systems into the main data warehouse.
- Worked with teams to integrate data from various sources, including **student performance data**, **exam scores**, and **feedback systems**, enabling better analysis and reporting.
- Developed custom **SQL queries** to retrieve and transform large sets of student data, improving the speed and accuracy of reporting and analytics for educational programs.
- Built **data validation** and **monitoring** frameworks to ensure the integrity of assessment data, flagging any discrepancies or errors in real-time.
- Collaborated with data scientists and product teams to understand business needs and translate them into scalable, efficient data solutions.
- Worked extensively with **PostgreSQL** and **MySQL** databases, ensuring data integrity and efficient querying for educational data applications.
- Contributed to building **data models** to track **student progress**, **test outcomes**, and **learning achievements**, improving reporting capabilities and supporting curriculum improvements.
- Created automated systems for **data extraction**, streamlining the process of pulling relevant educational data for reports and improving data efficiency.
- Optimized **data storage** and retrieval strategies, enhancing the overall performance of educational data systems and increasing throughput for real-time reporting.

Education

Ohio University

Bachelor's degree in computer science
Athens, OH

2008 – 2012